NutriCart Hub

1. Concept Overview

NutriCart Hub combines the strongest parts of the original Social Shopping Companion, Recipe-to-Cart, and Housemate Meal Coordination concepts into one physical system. It is designed as a shared household grocery-planning device with a detachable shopping unit. The system does not require a phone or a standalone app.

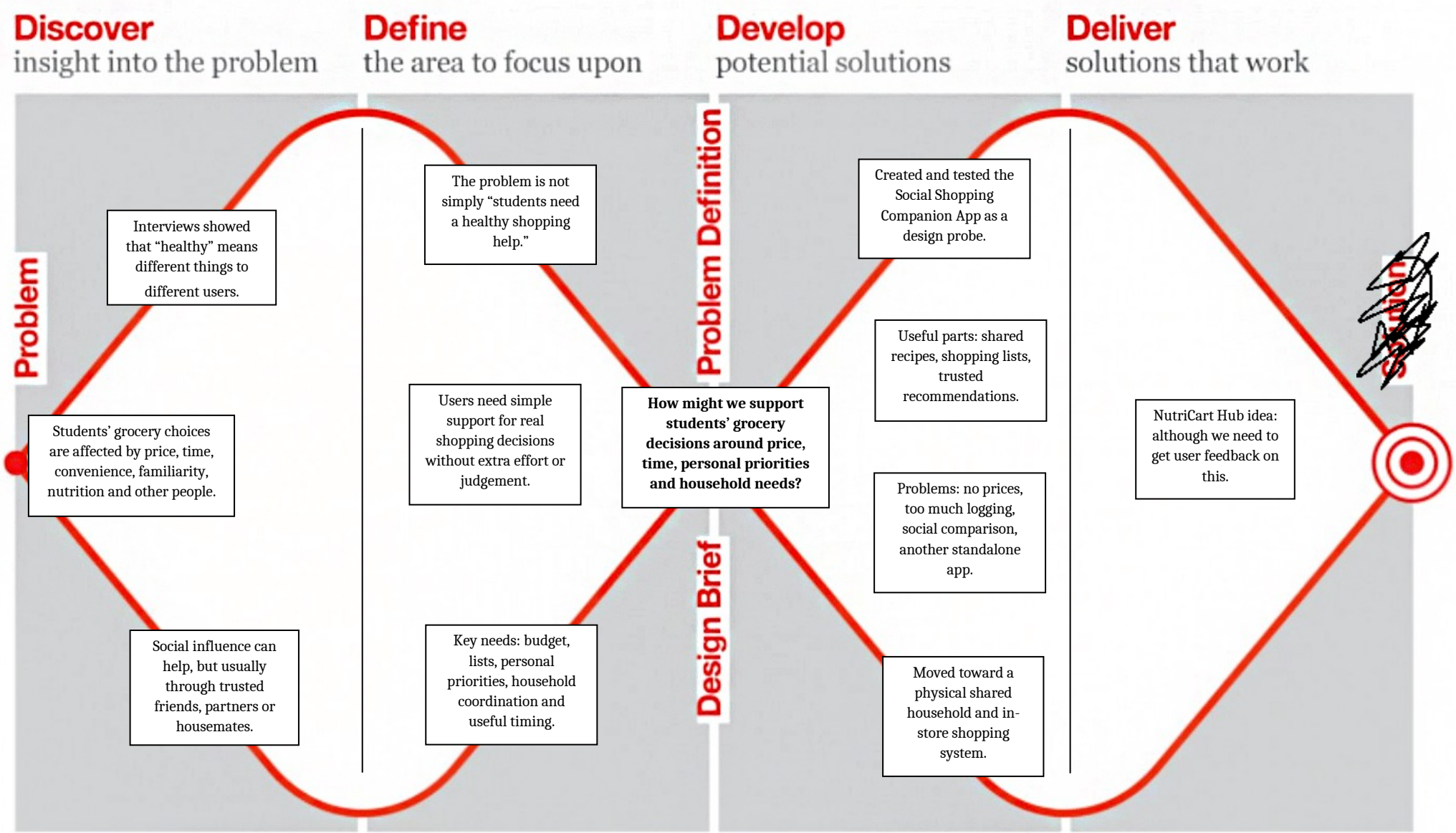
The larger Home Dock stays in the kitchen and is used for meal planning, shared grocery lists, household requests and deciding what needs to be bought. A smaller Cart Unit detaches from the Home Dock and goes to the supermarket, where it focuses only on the shopping list, budget, prices, product comparison and simple household recommendations.

2. Why the Concept Changed

The app prototype was useful because it showed which parts of the original idea people actually cared about. The feedback did not point towards simply making a better app. Instead, several patterns kept coming up:

- Price and specials strongly affect what students buy.
- Time pressure changes planned purchases and makes people choose quicker or easier options.
- People already rely on shopping lists and often want to get through the shop quickly.
- Friends and housemate recommendations can be useful when they come from people the shopper trusts.
- Group health goals and comparison were motivating for some people but uncomfortable or unnecessary for others.
- There was no consistent definition of what a healthy choice is. Different people cared about protein, sugar, fat, ingredients, processing, price, convenience or not much about health at all.
- Some people liked meal inspiration, but they wanted it to connect to the practical question of what they should actually buy.
- People did not want to constantly log food or spend a lot of time interacting with a system while shopping.

The new concept keeps social support, recipes and household coordination, but removes the idea that the system should judge whether someone is eating well. It supports the shopper with information and lets them decide what matters.


Discover
insight into the problem
Define
the area to focus upon
Develop
potential solutions
Deliver
solutions that work
Problem
Problem Definition
Design Brief
Solution
Interviews showed that "healthy" means different things to different users.
Students' grocery choices are affected by price, time, convenience, familiarity, nutrition and other people.
Social influence can help, but usually through trusted friends, partners or housemates.
The problem is not simply "students need a healthy shopping help."
Users need simple support for real shopping decisions without extra effort or judgement.
Key needs: budget, lists, personal priorities, household coordination and useful timing.
How might we support students' grocery decisions around price, time, personal priorities and household needs?
Created and tested the Social Shopping Companion App as a design probe.
Useful parts: shared recipes, shopping lists, trusted recommendations.
Problems: no prices, too much logging, social comparison, another standalone app.
Moved toward a physical shared household and in-store shopping system.
NutriCart Hub idea: although we need to get user feedback on this.

3. Combined concepts from previous pool of ideas

Original concept	What stays in NutriCart Hub
A. Social Shopping Companion	Trusted recommendations from housemates or friends, but no social feed or public leaderboard.
B. Recipe-to-Cart	A selected meal can be turned directly into a list of ingredients to buy.
C. Housemate Meal Coordination	Housemates can add groceries, suggest meals, leave requests and coordinate one shared shop.

4. Physical Components

Home Dock

The Home Dock stays on the fridge or kitchen bench. It is the planning and household coordination part of the system.

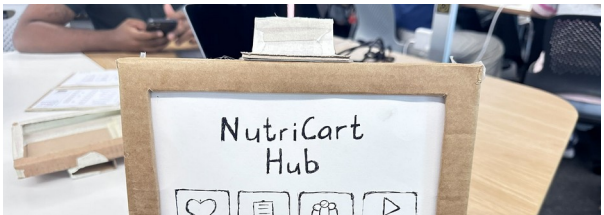
- Larger screen.
- Shared shopping list.
- Meal suggestions.
- Recipe-to-list conversion.
- Household requests.
- What is already at home.
- Rough shopping budget.
- Start Shop function that prepares the detachable unit.

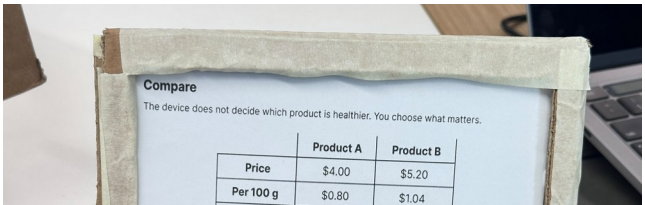
Cart Unit

The Cart Unit detaches from the Home Dock and clips onto a shopping trolley or basket. It is intentionally much simpler because the shopper may be rushed or focused on getting through the store.

- Shopping checklist.
- Running budget.
- Basic product scanner.
- Two-product comparison.
- Optional housemate recommendations.
- Deal alerts.
- Already-at-home reminders.
- Split Shop mode for two people shopping together.

5. Low Fidelity Physical Concept







6. Main Functionality

Feature	Function
Shared shopping list	Housemates can add items, quantities and who requested each item.
Meal suggestions	Housemates can suggest meals with practical details such as approximate cost, time and servings.
Recipe to shopping list	Choosing a meal creates an ingredient list and separates what is already at home from what needs to be bought.
Housemate requests	Short notes can be left for the shopper, such as preferred products or reminders.
Budget tracking	A rough household budget is set before leaving and the running total is shown during the shop.
Product comparison	Two products can be scanned and compared using price, unit price and selected nutrition information.
Deal alerts	A simple store deal can be shown when it is relevant to an item on the list.
Already at home	The device can warn the shopper that the household has some of the items left.
Optional recommendations	Trusted people can leave useful product suggestions without creating a social feed.
Split Shop	Two shoppers can divide the list and see which items each person is responsible for.

7. Full Interaction Flow

At home

1. Housemates add groceries they need.

2. Someone suggests a meal.

3. The household chooses the meal.

4. NutriCart converts the meal into ingredients.

5. The system checks which ingredients are already at home.

6. Missing ingredients are added to the shared shopping list.

7. Housemates leave any requests or product recommendations.

8. The shopper checks the list and budget.

9. The shopper presses Start Shop and detaches the Cart Unit.

At the supermarket

1. The Cart Unit shows a simple shopping checklist and budget.

2. The shopper tick items off as they go.

3. The running total updates when an item is added.

4. If the shopper is unsure between two products, they can use Compare.

5. Optional deal, household or already-at-home information appears only when relevant.

6. The shopper finishes the list and returns the Cart Unit to the Home Dock.

8. Home Dock Screens

Screen 1: Household Home

This is the main screen. It gives a quick summary of what the household is planning and what needs to be bought.

- Tonight's selected meal.
- Number of shopping items.
- Rough budget.
- Buttons for Meals, Shopping List, Household and Start Shop.

Screen 2: Shared Shopping List

This is one of the core parts of the concept. Each item can show who added it, the quantity and whether it came from a recipe or a direct request.

- Milk - added by Alex.
- Rice - added by you.
- Chicken - added by Jess.
- Capsicum - added automatically from a recipe.

Screen 3: Meal Board

The Meal Board is the practical version of the original social recommendation idea. Instead of a social feed, housemates leave meal ideas that can be used for the next shop.

- Meal name.
- Who suggested it.
- Approximate cost.
- Preparation time.
- Number of servings.
- Select button.

Meals should not be labelled simply as healthy or unhealthy. The idea is to give useful information and let the household decide whether the meal suits them.

Screen 4: Selected Recipe

Once a meal is selected, the device shows the ingredients and separates them into two groups: what is already at home and what still needs to be bought.

Screen 5: Household Requests

This is a simple way for social information to support the shop without becoming a social network.

- "If you see the cheap Greek yoghurt, grab one."
- "We still have rice, do not buy more."
- "Need something quick for lunch."

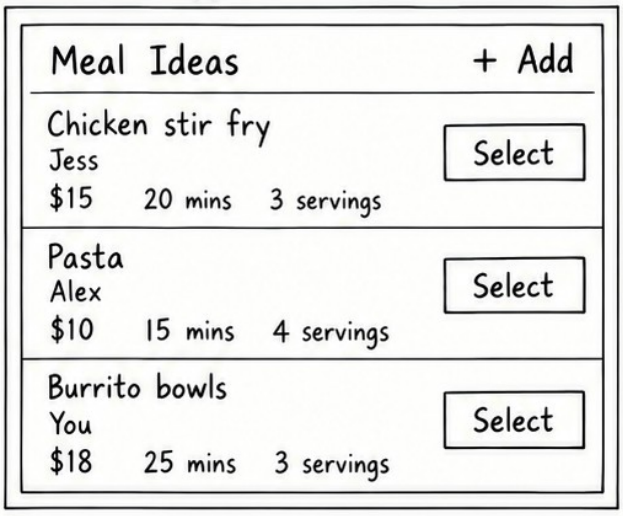
Screen 6: Ready to Shop

Before leaving, the device summarises the number of items, selected meals, budget and household requests. The shopper then detaches the Cart Unit.

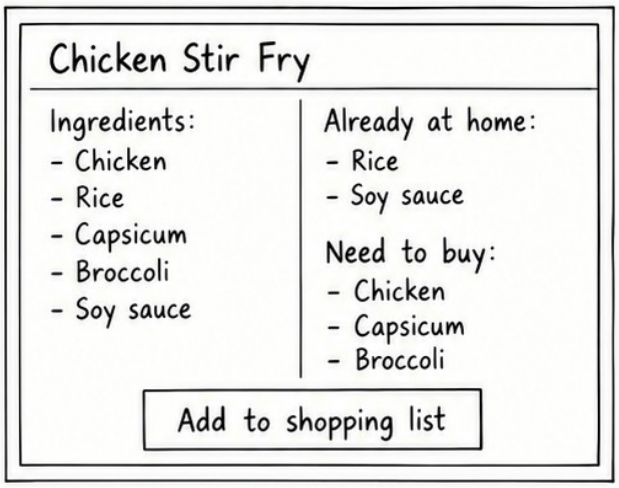
1. Household home



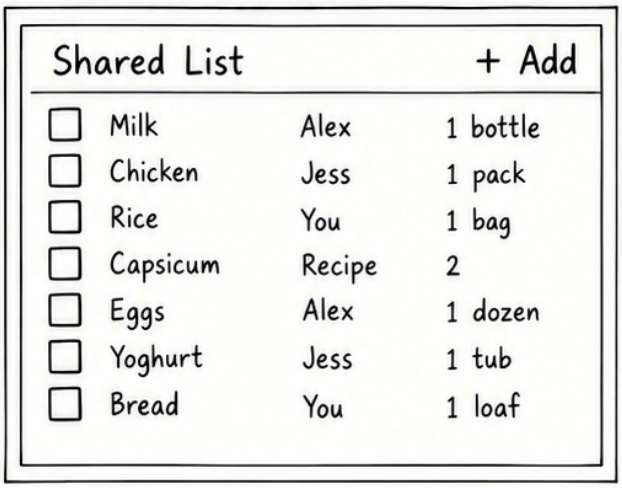
2. Meal suggestions



3. Selected recipe



4. Shared shopping list



5. Housemate requests



6. Ready to shop



9. Cart Unit Screens

Screen 1: Today's Shop

The main in-store screen is deliberately simple. It shows the shopping list and running budget with as little extra information as possible.

Screen 2: Item Checked Off

When an item is added, the device records the price, updates the running total and shows how many items are left.

Screen 3: Product Comparison

The shopper can scan two products when they are unsure which one to buy. The device shows information side by side, but it does not label either product as the best or healthiest option.

	Product A	Product B
Price	$4.00	$5.20
Price per 100 g	$0.80	$1.04
Protein	8 g	15 g
Sugar	9 g	4 g

The user still makes the decision. The purpose of the comparison is to make relevant information easier to see, not to decide what is healthy for them.

Screen 4: Housemate Recommendation

A recommendation only appears when it relates to something the shopper is already considering. It can be opened or ignored.

Example: "Get the cheaper Woolworths one. I used it last time and it was fine."

Screen 5: Deal Alert

If a relevant item is on special, the device can show the usual price and current price. The shopper can add it to the list or ignore it.

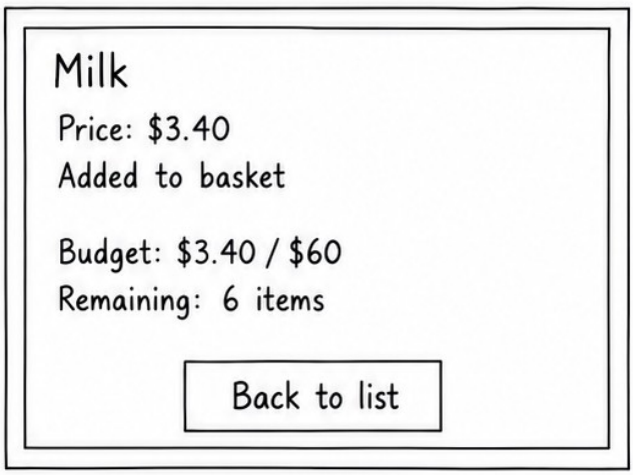
Screen 6: Already at Home

If the household has already marked that an item is still available at home, the device can warn the shopper before they buy another one.

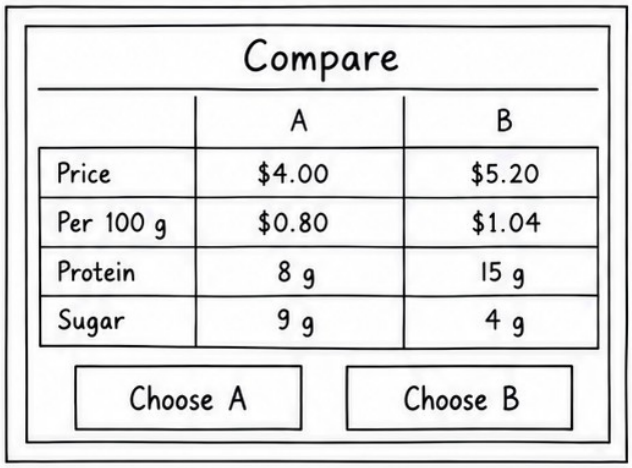
1. Shopping list



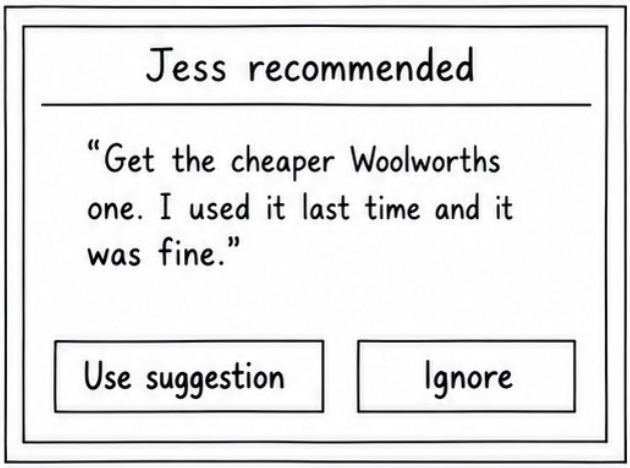
2. Item checked off



3. Product comparison



4. Housemate recommendation



5. Deal alert



6. Already at home



10. Optional Physical Priority Buttons

A low fidelity version could also test three large physical buttons on the Cart Unit:

- PRICE
- TIME
- NUTRITION

The shopper presses the one that matters most for that decision. The comparison screen then puts that information first. The purpose is not to assume that these three options are correct, but to test whether shoppers want a quick way to state what matters to them in the moment.

11. Split Shop Mode

For people shopping together, the list can be divided between two shoppers.

Shopper 1	Shopper 2
Chicken	Capsicum
Milk	Broccoli
Bread	Eggs
	Yoghurt

For a low fidelity test, the two devices do not need to sync electronically. The researcher can manually remove an item from the other person's paper screen when it is collected.

12. End of Shop

At the end, the Cart Unit shows a simple summary:

- Items completed.
- Total spent.
- Budget remaining.
- Whether the selected meal ingredients are complete.
- Whether household requests were completed.

The summary should not score the shopper, count calories, or say whether they were good or bad.

13. What Was Removed From the App Prototype

Original feature	New decision
Friend feed	Replaced with occasional household recommendations.
Group health leaderboard	Removed.

Healthy meal targets	Removed.
Constant food logging	Removed.
Standalone app	Replaced with a Home Dock and detachable Cart Unit.
Generic friend recommendations	Limited to trusted household or friend input.
Recipe browsing	Kept, but tied directly to the shopping list.
Shopping list	Expanded into a shared household list.
Location notification	Replaced by the physical transition from Home Dock to Cart Unit.
Health score	Replaced with side-by-side product information.
No price information	Budget, prices and unit prices are now central.

15. Simple Prototype Testing Setup

The concept can be tested with a small fake grocery setup before taking it into a real shadowing session.

1. Give the participant the Home Dock and explain that they live with two housemates.

2. Ask them to look at the household list and meal suggestions.

3. Let them choose a meal and convert it into ingredients.

4. Let them review what is already at home and any household requests.

5. Have them press Start Shop and physically detach the Cart Unit.

6. Set up a few grocery items on a table as a fake store.

7. Give them a choice between two similar products and allow them to use Compare if they want.

8. Show an Already at Home reminder for one item.

9. Show a housemate recommendation for one item.

10. Show a deal alert for one item.

11. Watch which functions they use and which they ignore.

16. Main Assumptions to Test

- A shared household list is useful enough to justify a dedicated physical device.
- Meal planning before shopping helps people decide what to buy.
- Price and budget information are useful during the actual shopping trip.

- People prefer comparison information over a system deciding what is healthy for them.
- Trusted recommendations can be useful without creating a social feed.
- A detachable physical device can provide support without slowing the shopper down.
- Planning information and in-store information should be separated.
- Already-at-home reminders can reduce unnecessary duplicate purchases.
- People shopping together may benefit from splitting one shared list.

17. What to Observe in the Next Test

- Which Home Dock page is the most useful to the user.

- Is the design of the product intuitive?

- Whether they what to use the meal suggestion.

- Whether they use the budget display while choosing products.

- Whether they choose to use the comparison function without being told to.

- Which comparison information they look at first.

- Whether a recommendation from a housemate changes their choice.

- Whether the recommendation feels useful, intrusive, or irrelevant.

- Whether an Already at Home reminder changes the purchase.

- Whether deal information changes the purchase.

- Whether the physical Cart Unit gets in the way (is it too much of a hassle/ what if they dont want to use a shopping cart).

- Whether they ask for information that the prototype does not currently show.